Marcus Fu

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PROFESSIONAL SUMMARY

Motivated 4th year Electrical Engineering student with strong programming skills and a focus on electronic systems and communication networks for power and automation applications. Experienced in designing, integrating, and testing complex hardware and software solutions for applied research and simulation projects.

TECHNICAL SKILLS

Equipment: Oscilloscopes | Logic Analyzer | Spectrum Analyzer | Multimeter | AD2 | PLC
Programming: C | C++ | Python | VBA | System Verilog | SCL | Ladder Logic
Software: MS Office Suite | CISCO Packet Tracer | Siemens TIA Portal | SOLIDWORKS | KiCad | MATLAB | DeltaV

EDUCATION

Bachelor of Engineering in Electrical Engineering 2022 - Present
British Columbia Institute of Technology - Burnaby, British Columbia
Coursework: Communication Networks | Digital System Design | Signal Processing and Filters | Analog & Digital Communication | PLC | Sensors for Measurement & Ctrl

WORK EXPERIENCE

FSA Student Researcher May 2025 - Ongoing
Smart Microgrid Applied Research Team – Burnaby, British Columbia
- Implemented IEC 61850 GOOSE and Sampled Values communication in C++ to enable Python-based AI control of a Real Time Digital Simulator (RTDS).
- Developed web-based Digital Twin environments with online and offline support, using Modbus and Excel spreadsheets for bidirectional data exchange.
- Authored technical manuals for integrating Digital Twins into multiple applications and for a self developed IEC 61850 to RTDS communication library.
- Created 3D Digital Twin models in SketchUp and MicroStation to support visualization and monitoring of BCIT's OASIS smart microgrid system.

Warehouse Associate August – September 2022, 2023
ARITZIA Warehouse Sale – Canada Place, Vancouver, British Columbia
- Managed apparel processing and inventory organization while assisting customers on the sales floor.
- Transported and staged heavy goods using on-site equipment such as pallet jacks, following safety procedures.
- Maintained productivity and attention to detail while performing repetitive tasks for extended periods.

COMPETITIONS

Western Engineering Competition (WEC) – *R&D / Sr. Programmer* Jan 2025
- Competed against the top 9 teams in Western Canada in the Programming category.
- Developed a MATLAB optimization algorithm to determine fire hall placements for urban expansion, balancing facility costs, computational efficiency, and long-term land development impacts.
- Collaborated with teammates to evaluate multiple urban planning solutions under strict time constraints.

BCIT Engineering Competition (BEC) – *Sensor Calibration and Integration* Nov 2024
- Competed in the Senior Design category to develop a transport vehicle intended to assist injured individuals.
- Designed and implemented sensor logic for IR and ultrasonic detection using an Arduino Uno (R3).

PROJECTS

Sep 2025 **Supercritical CO_2 Battery Recycling (Ongoing Capstone)**
- Coordinated project planning and scheduling to meet deliverables while maintaining technical and goal alignment across multidisciplinary teams.
- Currently designing P&IDs and control strategies for the recycling process using DeltaV.

May 2024 **Radio-Controlled Car, Integrating Autonomous and Manual Control**
- Utilized SOLIDWORKS and KiCad to design the chassis a single-layered PCB.
- Programmed autonomous driving/firing mode using image processing algorithms to detect ArUco targets.
- Established TCP/IP communication to provide real-time viewing from the mounted camera.